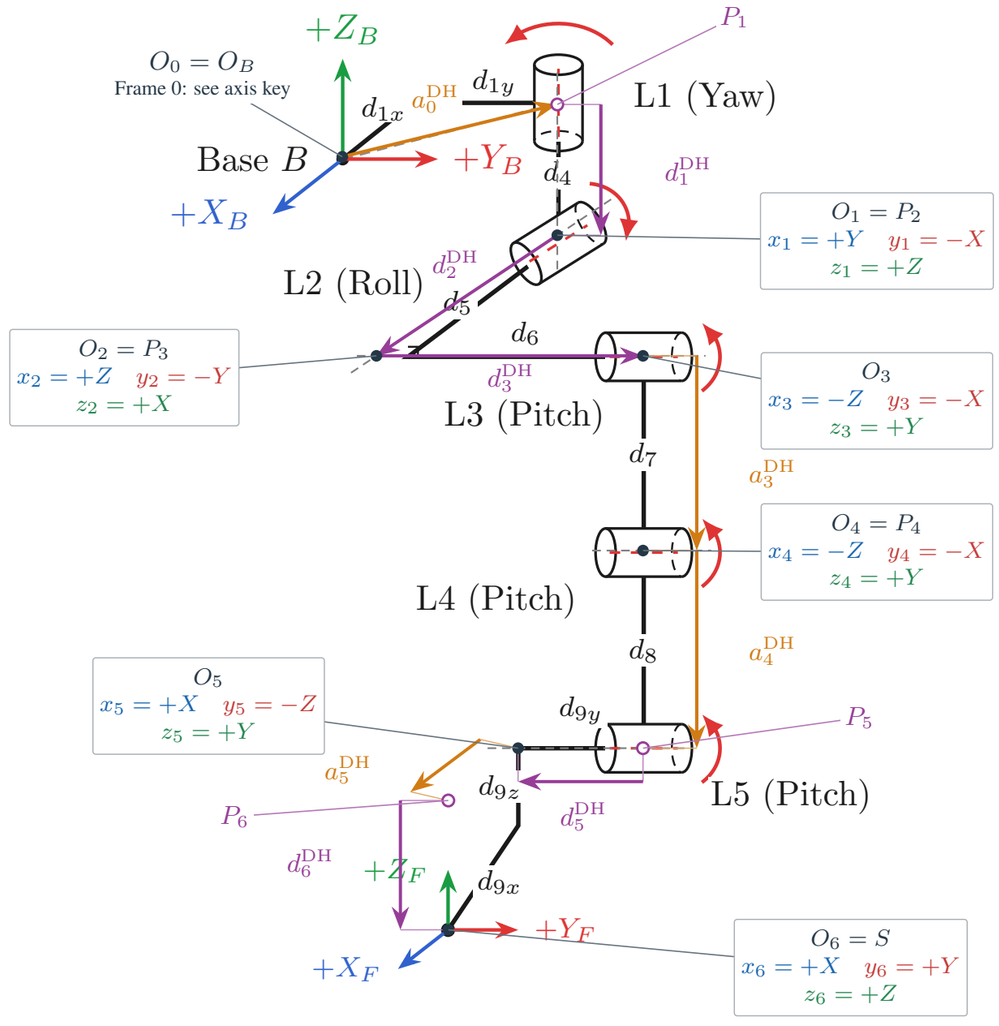
Left leg: D-H origins, construction distances and axes

Original assignment Fig. 2 retained underneath. Added labels use O_i, P_i, a_{i-1}^{DH} and d_i^{DH}; all listed lengths are in mm.



Filled dots: selected D-H origins. Open dots: intermediate P_i.
Boxed XYZ keys give directions in base B, not page directions.
Arrow paths show construction order on the original schematic.

Read each row as two translations

$$O_{i-1} \xrightarrow{a_{i-1}^{\mathrm{DH}}} \text{along } +x_{i-1} \quad P_i \xrightarrow{d_i^{\mathrm{DH}}} \text{along } +z_i \quad O_i.$$

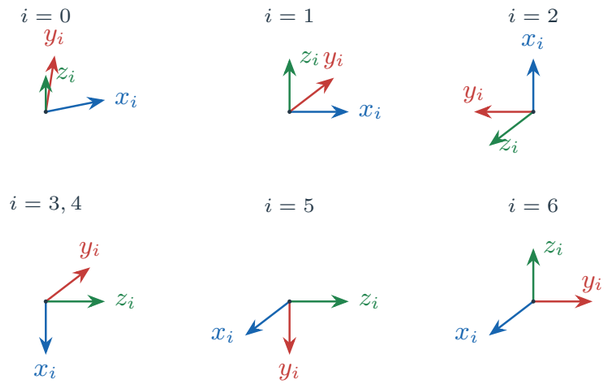
Negative values mean travel opposite to the stated positive axis.

Row	Route	a_{i-1}^{DH}	d_i^{DH}
1	$O_0 \to P_1 \to O_1$	$+39.8246$ along x_0	-0.0741 along $+Z$
2	$O_1 = P_2 \to O_2$	0	-16.060 along $+X$
3	$O_2 = P_3 \to O_3$	0	$+74.150$ along $+Y$
4	$O_3 \to P_4 = O_4$	$+78.650$ along $-Z$	0
5	$O_4 \to P_5 \to O_5$	$+78.650$ along $-Z$	-19.750 along $+Y$
6	$O_5 \to P_6 \to O_6$	$+0.500$ along $+X$	-36.225 along $+Z$

Positive coordinate axes at every origin (zero pose)

Frame	$+x_i$	$+y_i$	$+z_i$
0	$-sX + cY$	$-cX - sY$	$+Z$
1	$+Y$	$-X$	$+Z$
2	$+Z$	$-Y$	$+X$
3, 4	$-Z$	$-X$	$+Y$
5	$+X$	$-Z$	$+Y$
6 (site)	$+X$	$+Y$	$+Z$

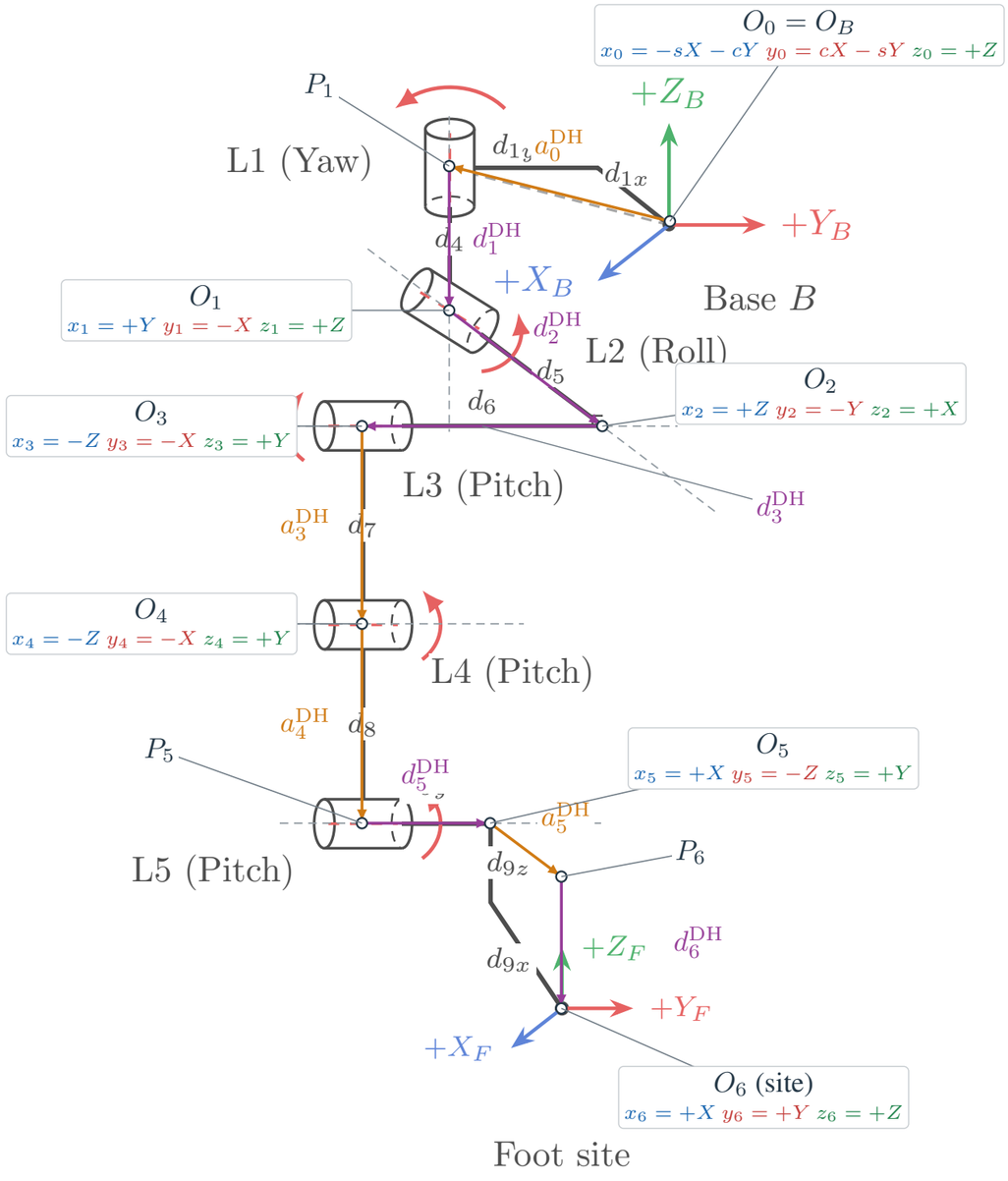
X, Y, Z denote the base axes; $s = 19/\sqrt{19^2 + 35^2}$, $c = 35/\sqrt{19^2 + 35^2}$. All frames satisfy $x_i \times y_i = z_i$.



Origin placement. O_1 is the yaw/roll axis intersection; O_2 is the roll/hip-pitch intersection. They are not the drawn cylinder centres. P_1 is a schematic yaw-axis reference at base height. For row 6, the construction takes the small a_5 step first, then d_6.
Source drawing versus signed geometry. The original is not to scale and its joint symbols do not share a calibrated 3-D projection. Gray joint-axis lines on the drawing are *unoriented*; the table and triads define the exact positive directions. Original labels $d_4, d_5, \ldots$ are handout dimension names, not D-H row numbers. In particular, $d_1^{\mathrm{DH}} = -0.0741$ is the selected origin's net height, not the CAD yaw-to-roll anchor separation.

Right leg: origins, D-H distances and axes

Direct annotations on the original assignment drawing. Zero pose; distances in mm.



Orange: a_{i-1}^{DH} Purple: d_i^{DH}

Original black d labels are handout dimensions. Coloured superscript DH labels are table parameters.

Read each row as a two-step path

$$O_{i-1} \xrightarrow[+x_{i-1}]{a_{i-1}} P_i \xrightarrow[+z_i]{d_i} O_i$$

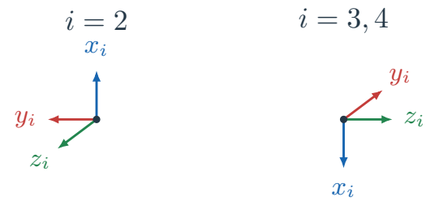
A negative value means travel opposite to that positive axis.

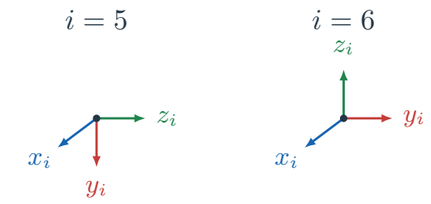
i	a_{i-1}	d_i	axis for a	axis for d
1	39.8246	−0.0741	$-sX - cY$	$+Z$
2	0	−16.06	$+Y$	$+X$
3	0	−74.15	$+Z$	$+Y$
4	78.65	0	$-Z$	$+Y$
5	78.65	17.35	$-Z$	$+Y$
6	0.5	−36.225	$+X$	$+Z$

Positive-axis triads

Translated from their O_i







$a_1 = a_2 = 0$; $d_4 = 0$.
$P_2 = O_1$, $P_3 = O_2$, $P_4 = O_4$.

Full frame axes, expressed in base B

frame	$+x_i$	$+y_i$	$+z_i$
0	$-sX - cY$	$cX - sY$	$+Z$
1	$+Y$	$-X$	$+Z$
2	$+Z$	$-Y$	$+X$
3	$-Z$	$-X$	$+Y$
4	$-Z$	$-X$	$+Y$
5	$+X$	$-Z$	$+Y$
6	$+X$	$+Y$	$+Z$

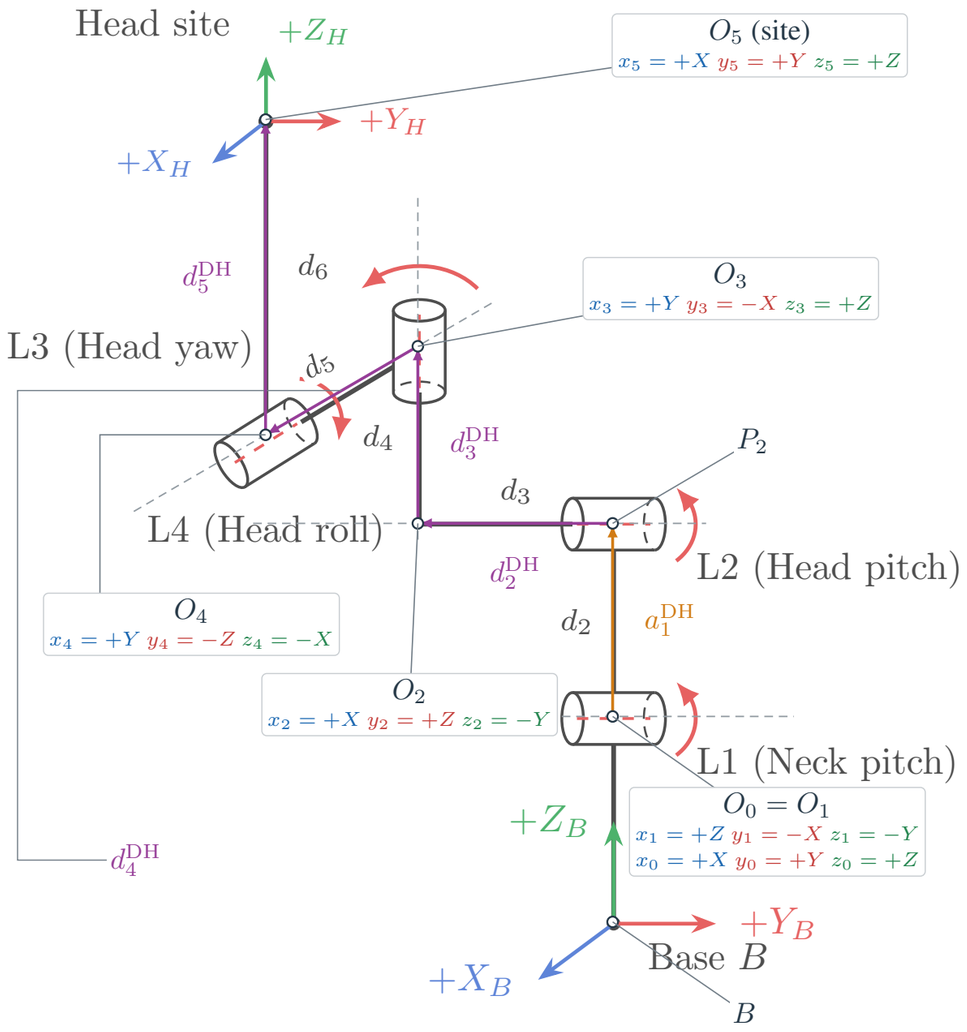
$s = \sin\beta$, $c = \cos\beta$, $\beta = \mathrm{atan2}(19, 35)$.

O_1: yaw/roll axis intersection. O_2: roll/hip-pitch axis intersection. P_5 and O_5 share the ankle axis. P_6 is the auxiliary corner for the D-H translation order. $O_0 = B$. The first vertical segment ends at O_1; $d_1 = -0.0741$ mm. Cylinder centres are not assumed to be D-H origins.

Original: Assignment 1, Fig. 3. Schematic, not to scale; perspective is illustrative. Exact signs and directions are given in the tables. Markers are selected D-H origins, not CAD joint centres.

Head and neck: origins, D-H distances and axes

Direct annotations on the original assignment drawing. Zero pose; distances in mm.



Orange: a_{i-1}^{DH} Purple: d_i^{DH}

Original black d labels are handout dimensions. Coloured superscript DH labels are table parameters.

Read each row as a two-step path

$$O_{i-1} \xrightarrow[+x_{i-1}]{a_{i-1}} P_i \xrightarrow[+z_i]{d_i} O_i$$

A negative value means travel opposite to that positive axis.

i	a_{i-1}	d_i	axis for a	axis for d
1	0	0	$+X$	$-Y$
2	66	19.05	$+Z$	$-Y$
3	0	37.85	$+X$	$+Z$
4	0	-5	$+Y$	$-X$
5	0	42.45	$+Y$	$+Z$

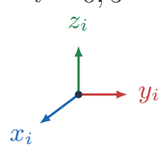
$a_0 = a_2 = a_3 = a_4 = 0$; $d_1 = 0$.
$P_1 = O_0 = O_1$, $P_3 = O_2$, $P_4 = O_3$, $P_5 = O_4$.

Full frame axes, expressed in base B

frame	$+x_i$	$+y_i$	$+z_i$
0	$+X$	$+Y$	$+Z$
1	$+Z$	$-X$	$-Y$
2	$+X$	$+Z$	$-Y$
3	$+Y$	$-X$	$+Z$
4	$+Y$	$-Z$	$-X$
5	$+X$	$+Y$	$+Z$

Positive-axis triads
Translated from their O_i

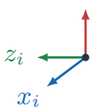
$i = 0, 5$



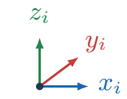
$i = 1$



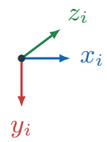
$i = 2$



$i = 3$



$i = 4$



$O_0 = O_1$ at the neck axis; their orientations differ. P_2 is the pitch-axis point vertically above O_1. O_2 is shifted along the head-pitch axis. O_3 and O_4 lie on extended yaw and roll axes. The base-to-neck transform ${}^{B}T_0$ is separate. The handout's $d_1 = 92.021$ mm is not the D-H $d_1 = 0$.

Original: Assignment 1, Fig. 4. Schematic, not to scale; perspective is illustrative. Exact signs and directions are given in the tables. Markers are selected D-H origins, not CAD joint centres.